

FUNDACIÓN FULGOR

CMOS Course

Homework N°3

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1 Introduction

This report analyzes the *DC* operating point and transient response of a *Common Source CMOS* amplifier with Biasing shown in Fig. 1.

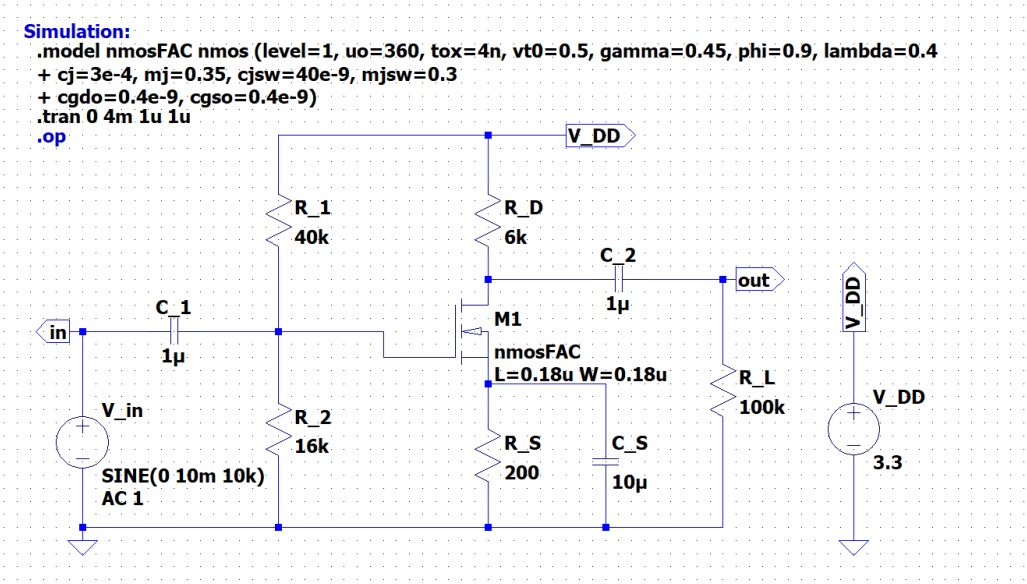


Figure 1: Circuit proposed to analyze.

Using the provided $W = L = 0.18\ [\mu m]$ *CMOS* model, theoretical hand calculations are compared with *LTspice* simulations to verify proper saturation bias and evaluate the dynamic mid-band voltage gain.

2 DC Operating Point Analysis

The objective of this section is to ensure the *MOSFET* is properly biased in the saturation region (linear amplification zone) using the provided *nmosFAC* model for $W = L = 0.18\ [\mu m]$.

2.1 Theoretical Calculations

Assuming the gate current is negligible ($I_G = 0$), the gate voltage V_G is determined by the resistive voltage divider formed by R_2 and R_3:

$$V_G = \frac{V_{DD} \cdot R_2}{R_1 + R_2} = \frac{3.3 \cdot 16k}{16k + 40k} \approx 0.9429\,[V] \tag{1}$$

Based on the *CMOS* model for the transistor provided shown below:

```
.model nmosFAC nmos (level=1, uo=360, tox=4n, vt0=0.5, gamma=0.45, phi=0.9,
+ lambda=0.4, cj=3e-4, mj=0.35, cjsw=40e-9, mjsw=0.3
+ cgdo=0.4e-9, cgso=0.4e-9)
```

We can see that the threshold voltage is equal to: $V_{Th0} = 0.5\,[V]$. Therefore, the source voltage V_S depends on the drain current I_D and the source degradation resistor R_S:

$$V_S = I_D \cdot R_S = I_D \cdot (200) \tag{2}$$

For the transistor to be turned on, the gate-source voltage must exceed the threshold voltage, this is: $V_{GS} = V_G - V_S > V_{Th}$. The drain current in saturation (ignoring channel-length modulation, this is: $\lambda = 0$ for the initial hand-calculation estimate) is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}((V_G - V_S) - V_{Th0})^2 \tag{3}$$

Replacing in Eq. 3 the value for V_{Th0} and the one for V_G obtained in Eq. 1, we have:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(0.9429 - I_D(200) - 0.5)^2 \tag{4}$$

From the *CMOS* model, we can also note that the oxide thickness is $t_{ox} = 4\,[nm]$. Then, we know that the value of C_{ox} is given by Eq. 5.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \tag{5}$$

where ϵ_{ox} is the relative permittivity of SiO_2 and is equal to:

$$\epsilon_{ox} = \epsilon_r \cdot \epsilon_0 = 3.9 \cdot 8.854x10^{-14} \approx 3.453x10^{-13}\ \left[\frac{F}{cm}\right] \tag{6}$$

Therefore, we have that combining Eq. 5 and Eq. 6, we have:

$$C_{ox} = \frac{3.453x10^{-13}\ \left[\frac{F}{cm}\right]}{4x10^{-7}\,[cm]} \approx 8.63x10^{-7}\ \left[\frac{F}{cm^2}\right] \tag{7}$$

Given that the electron mobility is equal to: $\mu_n = 360\,[\frac{cm^2}{Vs}]$ and that the *CMOS* model has a: $W = L = 0.18\ [\mu m]$, we can replace these values and the one obtained in Eq. 7, into Eq. 4 (recall that we are initially ignoring channel-length modulation, this is: $\lambda = 0$), so that we have:

$$I_D = \frac{1}{2} \cdot \left(360\,\frac{cm^2}{Vs}\right) \cdot \left(8.63x10^{-7}\,\frac{F}{cm^2}\right) \cdot \left(\frac{0.18\,\mu m}{0.18\,\mu m}\right) \cdot (0.4429 - 200 \cdot I_D)^2 \tag{8}$$

$$I_D \approx 155.34x10^{-6}(0.4429^2 - 2 \cdot 0.4429 \cdot 200 \cdot I_D + 40000 \cdot I_D^2) \tag{9}$$

$$I_D = 30.48x10^{-6} - 0.02753 \cdot I_D + 6.216 \cdot I_D^2 \tag{10}$$

Rearranging the expression obtained in Eq. 10 into the standard quadratic form:

$$6.216 \cdot I_D^2 - 1.02753 \cdot I_D + 30.48x10^{-6} = 0 \tag{11}$$

Now we can solve for I_D using the quadratic formula, where we obtain the following two solutions:

$$\begin{cases} I_{D1} = 0.1652\,[A] \\ I_{D2} = 29.668\,[\mu A] \end{cases} \tag{12}$$

We note that from Eq. 12 that we have two possible solutions for I_D yet only the smaller value is physically valid. The larger root would result in an excessive voltage drop across the source resistor (R_S), forcing the gate-source voltage (V_{GS}) below the threshold voltage (V_{Th}) and violating the active conduction assumption. Therefore, the lower current value is strictly chosen to satisfy the physical turn-on condition $V_{GS} > V_{Th}$.

Now, with the drain current established, the preliminary drain-source voltage V_{DS} is:

$$V_{DS} = V_{DD} - I_D \cdot (R_D + R_S) = 3.3 - 29.668\mu \cdot (6200) \approx 3.116\,[V] \tag{13}$$

Notice that the provided physical parameters describe a short-channel device and even the model for the *CMOS* transistor includes a non zero channel-length modulation parameter (this is: $\lambda = 0.4$) which significantly increases the current. If we apply the corresponding correction factor, we have:

$$I_{D;real} = I_D \cdot (1 + \lambda \cdot V_{DS}) = 29.668x10^{-6} \cdot (1 + 0.4 \cdot 3.116) \approx 66.64\,[\mu A] \tag{14}$$

Finally, we recalculate the true *DC* drain-source voltage using the corrected current:

$$V_{DS;real} = V_{DD} - I_{D;real} \cdot (R_D + R_S) = 3.3 - 66.64x10^{-6}(6200) \approx 2.886\,[V] \tag{15}$$

2.2 *LTspice* Simulation Results

The circuit was implemented in *LTspice* using the *CMOS* model given. A *DC* operating point simulation was executed using the `.op` directive where we obtained he following results:

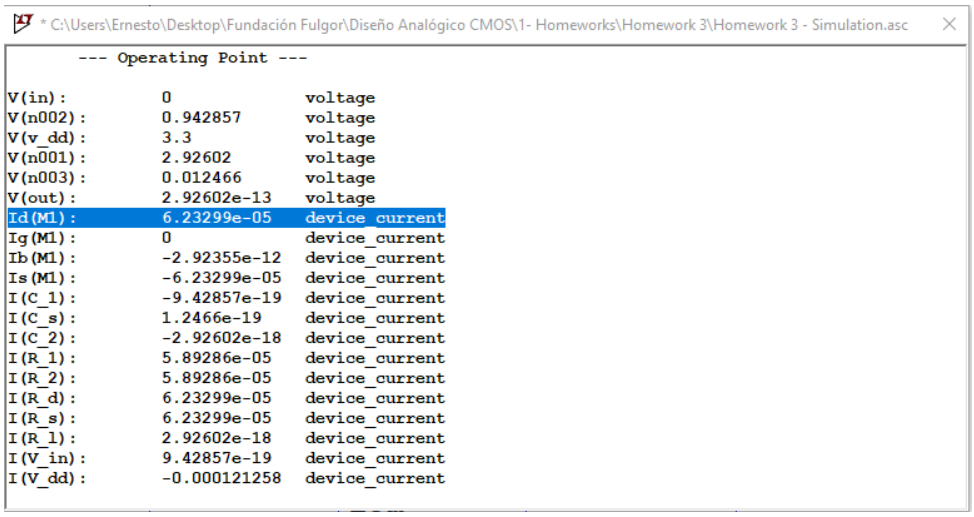


Figure 2: Results of `.op` simulation in *LTspice*

With the results calculated in Eq. 1, Eq. 14, Eq. 15 and the results given by the simulation in Fig. 2, we can make a comparison table of the simulation versus calculation results:

Parameter	**Theoretical Estimate**	**LTspice Simulation**
V_G	$0.9429\,[V]$	$0.942857\,[V]$
I_D	$66.64\,[\mu A]$	$62.3299\,[\mu A]$
V_{DS}	$2.886\,[V]$	$2.913554\,[V]$

Table 1: Comparison of theoretical hand calculations and *LTspice DC* simulation.

We can see that the values calculated are practically identical to the ones simulated, therefore validating the model and the equations used in the subsection 2.1. Note that the small discrepancies between theoretical and simulated values are mainly attributed to the body effect (γ) not being manually calculated, which slightly increases the threshold voltage as $V_S > 0$ that we can verify as $V_S \approx 12.46\,[mV]$ in Fig. 2.

3 Transient Response (Time-Domain Analysis)

The objective of this phase is to evaluate the dynamic processing and amplification of a small, time-varying signal.

3.1 Simulation Setup

- **Source Configuration:** The input voltage source (V_{in}) was configured as a sine wave (`SINE`) with no *DC* offset, a peak amplitude of $10\,[mV]$, and a frequency of $10\,[kHz]$.

- **Transient Analysis:** The simulation was executed using the `.tran 0 4m 1u 1u` command. Because the period of a $10\,[kHz]$ signal is $0.1\,[ms]$, a $4\,[ms]$ simulation time captures several complete cycles to properly observe the steady-state dynamic behavior.

3.2 Measurements and Voltage Gain

Both the input signal at the gate and the amplified output signal were plotted simultaneously and we can see the results of this simulation in Fig. 3.

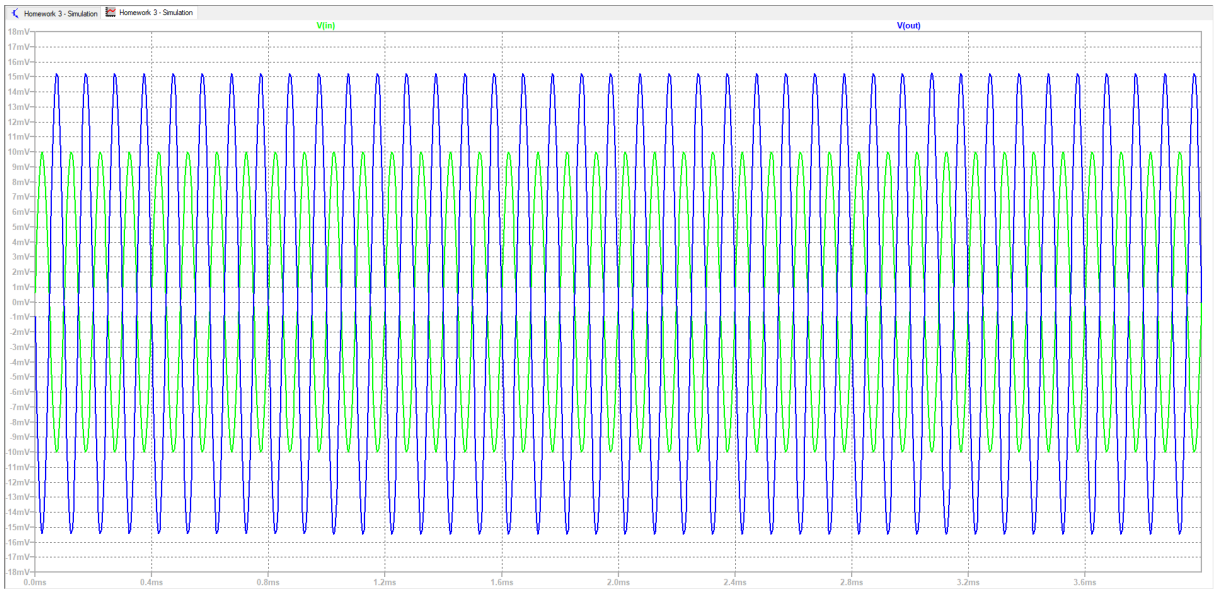


Figure 3: Results of `.tran` simulation in *LTspice*, where we see in green the input voltage (v_{in}) and in blue the output voltage (v_{out}).

We can see from the y-axis, that:

- Input peak-to-peak voltage: $V_{in,pp} = 20\,[mV]$
- Output peak-to-peak voltage: $V_{out,pp} = 30\,[mV]$

Therefore, the mid-band voltage gain (A_v) of the amplifier is calculated as the ratio of the output to the input peak-to-peak voltage, this is:

$$A_v = \frac{V_{out,pp}}{V_{in,pp}} = \frac{30\,[mV]}{20\,[mV]} = 1.5\,\left[\frac{V}{V}\right] \tag{16}$$

Note that the result obtained in Eq. 16 shows an exceptionally low simulated mid-band voltage gain. This is not an error but rather a direct consequence of the dimensions of the device (where $W = L = 0.18\,[\mu m]$): An aspect ratio of $W/L = 1$ restricts the drain current to approximately $62\,[\mu A]$, severely limiting the transconductance (g_m) and since the voltage gain is proportional to g_m, the amplifier cannot provide significant amplification under these sizing constraints.

4 Conclusion

This report analyzed the *DC* operating point and transient response of a Common Source *CMOS* amplifier. The section 2 demonstrated that theoretical calculations, especially when including the channel-length modulation parameter (λ), closely match the *LTspice* simulations; where as section 3 confirmed a proper linear operation but revealed a marginal voltage gain ($1.5\,[V/V]$). This highlights a key principle in analog *CMOS* design: while correct *DC* biasing is necessary, proper device sizing (W/L) is strictly required to achieve the desired amplification.